INT202 Complexity of Algorithms

Tutorial 9

Q1: Consider the following graph $G=(V,E)$:

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V = {a, b, c, d, e, f}
E = {(a,b), (b,c), (c,d), (d,e), (e,f), (f,a), (a,c), (b,e)}
```

1) Run the APPROX-VERTEX-COVER algorithm (iteratively pick an uncovered edge and add both endpoints to the cover). Assuming edges are chosen in the order shown in E, write down the resulting vertex cover C and its size.
2) Find a minimum vertex cover of this graph (by reasoning or trial), and give its size OPT.
3) Compute the approximation ratio of this execution, and state whether it exceeds 2.

Q2: In a complete graph G (a type of undirected graph in which every pair of distinct vertices is connected by a unique edge) satisfying the triangle inequality, vertices are A, B, C, D with the following edge weights:

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AB = 1, AC = 4, AD = 3, BC = 2, BD = 5, CD = 1
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1) Run Prim's algorithm starting from A to find a minimum spanning tree T. List the edges in T and its total weight.
2) Perform a preorder traversal on T starting from A. Write down the order in which vertices are visited, and construct a Hamiltonian cycle H from that order.
3) Compute the total weight of H and compare it with the optimal TSP tour.